

# Engineering Physics (2025)

Course code 25PY101-S2

## Module 1 Unit 2: Quantum mechanics

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# M1U2 Plan

Introduction to QM

Dual nature of radiation

de Broglie's concept of matter waves

Heisenberg's uncertainty principle

Schrödinger's time dependent wave equation

Particle in a 1D box

Quantum dots

Finite Potential well – Quantum tunneling

# Introduction to QM

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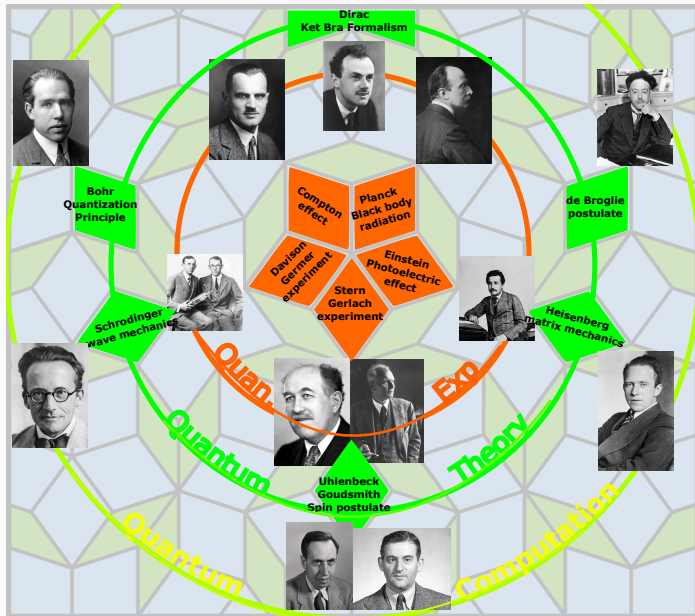
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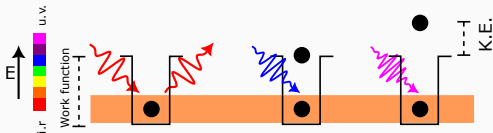
# Quantum mechanics jigsaw puzzle



# Photo-electric effect

## Definition

Light of **suitable** frequency leads to **instantaneous** emission of electron from a metal surface.



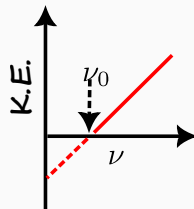
A.

Einstein

- Discovered by Albert Einstein in 1905.
- The emitted electron is called the **photo-electron**.
- The minimum energy required to emit an electron is a material property and is called the **work function** of the metal.

# Classical explanation of photo-electric effect

- Light is a wave with energy given by its intensity.
- Therefore, even low-frequency light (if sufficiently intense) should eventually provide enough energy to eject electrons — though perhaps after some time delay.
- But it is observed that
  - Emission of electrons is instantaneous – even for low intensity light as long as frequency of light is above a certain threshold  $\nu_0$ .
  - No electrons are emitted if the frequency is below the threshold, no matter how intense the light is.
  - The kinetic energy of emitted electrons depends on the frequency of light, not its intensity.



## Key Insight

Classical electromagnetic theory cannot explain photoelectric effect!

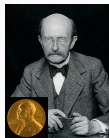


# Quantum mechanical explanation of Photoelectric effect

- Albert Einstein borrowed the idea of **quantum**<sup>1</sup> from Max Planck's theory of blackbody radiation and applied to photoelectric effect.

## Planck's law

- Max Planck postulated that the light energy is absorbed or emitted by matter in is a packet of energy. The quantum of energy is related to the frequency  $\nu$  by  $E = h\nu$  where  $h$  is the **Planck's constant** given by  $6.625 \times 10^{-34}$  Js.
- The quantum of light energy is called **photon**.



Max  
Planck

## Estimate: Energy of infra-red photon

Wavelength of infra-red photon is  $1\text{ }\mu\text{m}$ .

[Hint:  $c = \nu\lambda$ ]



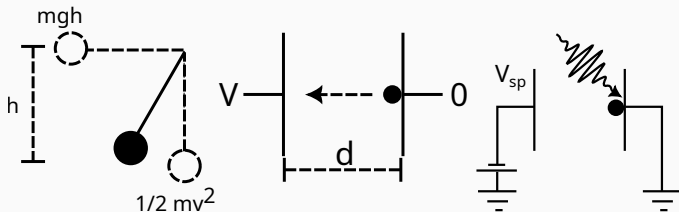
<sup>1</sup>“Quantum” is Latin for “a certain amount”. In the context of physics, the word quantum refers to a **packet**.



# Photoelectric effect: experiment

## Theorem

*An electron under potential difference of  $V$  gains kinetic energy of  $eV$ .*



## Proof.

- Electric field is  $E = \frac{V}{d}$ .
- Acceleration is  $|a| = \frac{eE}{m}$ .
- Initial velocity  $u = 0$ . Final velocity is  $v^2 = 2ad$ .
- Kinetic energy K.E.  $= \frac{1}{2}mv^2 = eV$ .

## Photoelectric effect: energy equation

- Photon with energy  $E = h\nu$  transfers energy to the electron, to overcome the work function  $\Phi$ . The rest of the energy is photoelectron's kinetic energy.

$$\text{K.E.} = h\nu - \Phi$$

- The photoelectrons generate photocurrent.
- In the experiment, the photo electron emitted is decelerated by applying a **stopping potential**  $V_{sp}$  till the photocurrent is zero. Thus,

$$\text{K.E.} = eV_{sp}$$

- Therefore, frequency is related to stopping potential as

$$eV_{sp} = h\nu - \Phi$$

- The plot of  $\nu(\rightarrow x)$  vs  $V_{sp}(\rightarrow y)$  is a straight line  $y = mx + c$  with slope  $m = h/e$  and y-intercept  $c = -\Phi$ .
- The x-intercept is the minimum frequency of photon to generate photo-current and is called the **threshold frequency**  $\nu_0 = \frac{\Phi}{h}$ .

## Photoelectric effect: momentum equation

- From Einstein's special theory of relativity, the momentum of a particle of mass  $m$  is related to its total energy  $E$  by

$$E^2 = p^2 c^2 + m^2 c^4$$

where  $c$  is the velocity of light.

- If the light is considered as a massless particle i.e.  $m_{\text{photon}} = 0$ , then

$$E = p_{\text{photon}} c \Rightarrow p_{\text{photon}} = \frac{E}{c}$$

- However, the velocity of light is related to its wavelength and frequency by

$$c = \nu \lambda$$

- Using the Planck's law for the energy of photon, the momentum is given by

$$p_{\text{photon}} = \frac{h}{\lambda}$$

# Photoelectric effect: problems

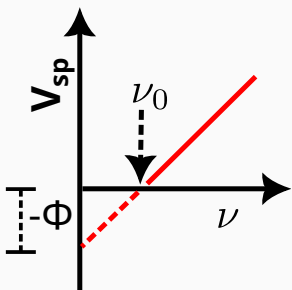
## Estimate: Energy of X-ray photon



Wavelength of X-ray photon is  $0.708 \text{ \AA}$ .

[Hint: (a).  $E \propto 1/\lambda$ . (b).  $E_{\lambda=1 \mu\text{m}} = 1.24 \text{ eV}$ ]

## Problem



*The work function  $\Phi$  of Au is  $4.90 \text{ eV}$ .*

- 1. A blue photon is incident on Au.  
Does it emit photo-electron? If yes,  
what is the stopping potential  
required to have zero photocurrent.  
[ $\lambda_{\text{blue}} = 460 \text{ nm}$ ]*
- 2. Determine the threshold frequency  $\nu_0$   
for photon emission of electrons for  
Au.*

*Repeat the above steps for Cs with work  
function  $1.90 \text{ eV}$ .*

# Applications of photo-electric effect: Smartphone



Image taken from Techbii.com. [Click here for link](#)

- Image sensors use Complementary Metal Oxide Semiconductor (CMOS) sensors
- Fingerprint sensor
- Proximity sensor (Infra-Red (IR) reflection and detection)
- Smoke detector

## Dual nature of radiation

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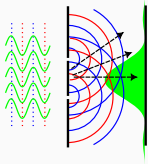
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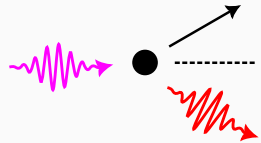
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# Radiation: Wave vs Particle



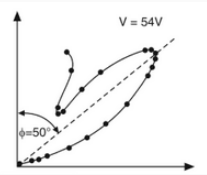
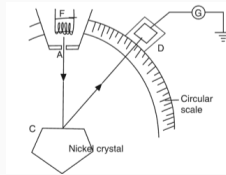
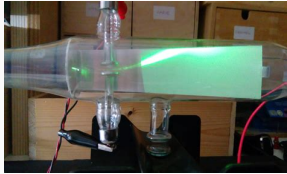
- Young's double slit experiment
- Discovered in 1801
- Light is a wave with amplitude, frequency and wavelength
- **Principle of superposition:** Waves superpose leading to constructive and destructive fringes.
- Animation for double slit experiment  
[Click]



- Compton effect
- Discovered in 1923.
- Light is a particle with momentum!
- **Conservation of momentum:** Light redshifts (wavelength increases) upon scattering off an electron.



# Matter: Particle vs Wave



(a)cathode ray tube, Davison-Germer experiment (b) and data (c).

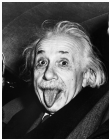
- Thomson's cathode ray tube experiment
- Discovered in 1897
- Electron is a particle
- Davisson and Germer experiment
- Discovered in 1927.
- Electron is a wave!

## Key Insight

The name of the game is light-matter interaction. And nature plays the game in a **symmetric** fashion!



# Summary of nature of light-matter interaction

|          | Wave                                                                                                                                                                                                                                                                                                                                                                                     | Particle                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Light    | <br>Young's double slit <sup>2</sup>                                                                                                                                                                                                                                                                    |    <br>Einstein's photo-electric effect<br>Compton's scattering effect |
| Electron |    <br>Davison's and Germer's diffraction experiment |  <br>Thomson's cathode ray tube experiment                                                                                                                                                                                                 |

<sup>2</sup>British polymath who has been described as “The Last Man Who Knew Everything” and disproved Newton's corpuscular theory of light.

## **de Broglie's concept of matter waves**

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Heisenberg's uncertainty principle

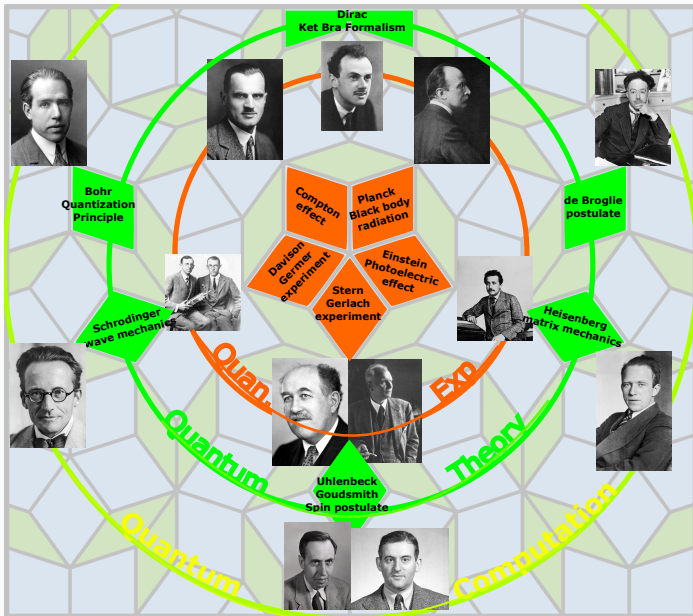
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# Quantum mechanics jigsaw puzzle



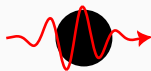
# Theories of quantum physics

- Max Planck's postulate of quantization of light energy absorbed or emitted by electron in quanta (plural for quantum)
- Louis de Broglie's postulate of existence of matter waves
- Werner Heisenberg's principle of uncertainty
- Erwin Schrödinger's wave equation

# Wave-Particle Duality

## de Broglie hypothesis

Any moving particle is associated with a wave. The waves associated with particles are called **de Broglie waves** or **matter waves**.



- Louis de Broglie in 1924 postulated the existence of **matter waves**.
- He suggested since light wave exhibits particle like behaviour, matter particles should be expected to show wave-like properties. This is the hypothesis of **wave-particle duality**.
- The wavelength  $\lambda$  of the matter wave is related to the momentum  $p$  of the matter particle by



$$\lambda = \frac{h}{p}$$

## Matter wave: properties

- Matter waves are due to motion of particle and are independent of charge. Therefore, they are neither electromagnetic waves nor acoustic waves. They are a new kind of waves.
- Can propagate/travel through vacuum and do not require any medium for propagation.
- Smaller the mass, longer the de Broglie wavelength.
- Smaller the velocity, longer the de Broglie wavelength.
- Velocity of matter waves depends on velocity of particle and is not a constant quantity.

**Estimate: de Broglie wavelength of thermal electron**

Estimate at room temperature.



**Estimate: de Broglie  $\lambda$  of macroscopic particle**



Cricket ball of mass 150 g is bowled at  $140 \text{ km h}^{-1}$ .





# Heisenberg's uncertainty principle

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# Heisenberg uncertainty principle

## Uncertainty principle/hypothesis

**Precise** values of position and momentum of a particle cannot be determined **simultaneously**. If the uncertainty in position is  $\Delta x$  and uncertainty in momentum is  $\Delta p$ , then

$$\Delta x \Delta p \geq \frac{\hbar}{2}, \quad \text{where} \quad \hbar = \frac{h}{2\pi}$$



where  $\hbar$  is the reduced/modified Planck's constant.

- Werner Heisenberg postulated the uncertainty principle in 1927.
- In classical mechanics, position and momentum are **deterministic**. In quantum mechanics, they are not deterministic and are associated with **uncertainties**.
- If position is measured with low uncertainty, then measurement of momentum has high uncertainty and vice versa.

## Uncertainty → Probability

- The uncertainty principle relates any pair of **conjugate** variables.
- Position – momentum is a pair of conjugate variables.
- Energy – time is also a conjugate variable pair.
- So the uncertainty in energy  $\Delta E$  is related to the uncertainty in time by  $\Delta t$  by

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

This is the second statement/corollary of Heisenberg's uncertainty principle.

- Due to the smallness of reduced Planck's constant, the uncertainty principle is significant for subatomic particles.
- The cause of uncertainty is **limitation of measurement**.

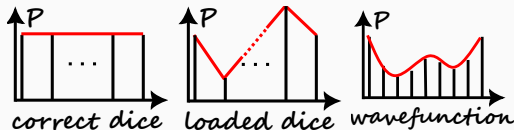
### Key Insight



Since the position cannot be measured with certainty, we determine the **probability** of finding an electron at a particular position.

# Probability: Mapping of Dice → position

- The way to deal with uncertainty is to talk in terms of probability.
- If a system has  $N$  events, the probability of event  $A$  is defined as number of times event  $A$  has occurred over total number of outcomes.
- A coin is a “2-sided polyhedron/dice”. The probability of coin having heads is  $\frac{1}{2}$ . Similarly for  $N$ -sided dice the probability is  $\frac{1}{N}$ .
- A “loaded” dice can different probability for each outcome.
- In the limit of  $\infty$ -sided dice, the probability of a outcome becomes a function.



# Uncertainty in function of variable

- If uncertainty in variable  $x$  is  $\Delta x$ , then the uncertainty in variable  $f(x)$  is given by

$$\Delta f = \frac{df}{dx} \Delta x$$

- Kinetic energy  $E$  is related to momentum  $p$  by

$$E = \frac{p^2}{2m}$$

- If uncertainty in momentum is  $\Delta p$ , then uncertainty in kinetic energy  $E$  is given by

$$\Delta E = \frac{dE}{dp} \Delta p = \frac{p}{m} \Delta p$$

## Problem

- *Uncertainty in position of electron is  $12 \text{ \AA}$ .*
- *Nominal energy of electron is  $16 \text{ eV}$ .*
- *Determine the uncertainty in momentum and kinetic energy*

# **Schrödinger's time dependent wave equation**

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# Schrödinger's wave equation

## Wave equation hypothesis

If the **wave function** of a particle of mass  $m$  is  $\Psi(x, t)$ , then it satisfies the **wave equation** given by

$$-\frac{\hbar^2}{2m} \cdot \frac{\partial^2 \Psi(x, t)}{\partial x^2} + V(x)\Psi(x, t) = j\hbar \frac{\partial \Psi(x, t)}{\partial t}$$

where  $V(x)$  is the potential function<sup>a</sup> and  $j$  is the imaginary constant  $\sqrt{-1}$ .



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<sup>a</sup>Potential energy and not electric potential difference

- Erwin Schrödinger postulated the wave equation in 1926 that incorporated principles of quanta introduced by Planck and wave-particle duality introduced by de Broglie.
- The wave function  $\Psi(x, t)$  describes the behaviour of particle
- $\Psi(x, t)$  can be a **complex** quantity.

## Analysis of the wave equation (W.E.)

$$-\frac{\hbar^2}{2m} \cdot \frac{\partial^2 \Psi(x, t)}{\partial x^2} + V(x)\Psi(x, t) = j\hbar \frac{\partial \Psi(x, t)}{\partial t}$$

---

1. The independent variables/quantities are position  $x$  and time  $t$  whereas the dependent variable is the wave function  $\Psi$ .

Therefore, wave function is a function of position and time

$$\Psi(x, t)$$

2. W.E. is a **partial** differential equation.

- A differential equation relates dependent variable  $y$  to the independent variable  $x$ . The relationship is given by derivatives  $\frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots$
- A partial differential equation relates dependent variable  $z$  to the independent variables  $x$  and  $y$ . The relationship is given by partial derivatives  $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \frac{\partial^2 z}{\partial x^2}, \dots$

## Analysis of the wave equation (W.E.) contd...

$$-\frac{\hbar^2}{2m} \cdot \frac{\partial^2 \Psi(x, t)}{\partial x^2} + V(x)\Psi(x, t) = j\hbar \frac{\partial \Psi(x, t)}{\partial t}$$

---

4. W.E. is a **second order** differential equation.
  - A first order differential equation has only first order derivative.
  - A second order differential equation has upto second order derivatives.
5. The potential energy  $V(x)$  is a parameter whereas  $\hbar, m, j$  are constants.

## Solution of the W.E. : Separation of variables

$$-\frac{\hbar^2}{2m} \cdot \frac{\partial^2 \Psi(x, t)}{\partial x^2} + V(x) \Psi(x, t) = j\hbar \frac{\partial \Psi(x, t)}{\partial t}$$

---

- The goal is to find the solution  $\Psi(x, t)$  for a given  $V(x)$ .
- We assume form of the solution as

$$\Psi(x, t) = \psi(x) \phi(t)$$

where  $\psi(x)$  is a function of the position only and  $\phi(t)$  is a function of the time only.

- Upon substitution of above form, W.E. reduces to

$$-\frac{\hbar^2}{2m} \phi(t) \frac{d^2 \psi(x)}{dx^2} + V(x) \psi(x) \phi(t) = j\hbar \psi(x) \frac{d \phi(t)}{dt}$$

## Solution of the W.E. : Separation of variables

$$\text{W.E.} \quad -\frac{\hbar^2}{2m} \phi(t) \frac{d^2 \psi(x)}{dx^2} + V(x) \psi(x) \phi(t) = j\hbar \psi(x) \frac{d\phi(t)}{dt}$$

$$\text{W.F.} \quad \Psi(x, t) = \psi(x) \phi(t)$$

- Divide the above equation by total wave function so that

$$-\frac{\hbar^2}{2m} \frac{1}{\psi(x)} \frac{d^2 \psi(x)}{dx^2} + V(x) = j\hbar \frac{1}{\phi(t)} \frac{d\phi(t)}{dt}$$

- Since L.H.S. is a function of  $x$  only and R.H.S. is a function of  $t$  only, each side must be equal to constant. Let us call this the **separation constant**  $\eta$ .

$$-\frac{\hbar^2}{2m} \frac{1}{\psi(x)} \frac{d^2 \psi(x)}{dx^2} + V(x) = \eta, \quad j\hbar \frac{1}{\phi(t)} \frac{d\phi(t)}{dt} = \eta$$

## Solution: Time dependent part of W.F

$$j\hbar \frac{1}{\phi(t)} \frac{d\phi(t)}{dt} = \eta$$

---

- The time dependent side of W.E. is then given by

$$\frac{d\phi(t)}{dt} = -j\frac{\eta}{\hbar}\phi(t) \quad \text{related to} \quad \frac{dy}{dx} = cx$$

- The solution is a sinusoidal wave given by

$$\phi(t) = e^{-j\left(\frac{\eta}{\hbar}\right)t} \quad \text{related to} \quad y = e^{-j\omega t}$$

with angular frequency  $\omega$  given by  $[\nu = \frac{\omega}{2\pi}]$

$$\omega = \frac{\eta}{\hbar}, \quad \Rightarrow \eta = \hbar\omega = \frac{h}{2\pi} \cdot 2\pi\nu = h\nu$$

- From Planck's law,  $E = h\nu$ . Therefore,

$$\eta = E, \quad \text{and} \quad \phi(t) = e^{-j\left(\frac{E}{\hbar}\right)t} = e^{-j\omega t}$$

## Solution: Time independent part of W.F

$$-\frac{\hbar^2}{2m} \frac{1}{\psi(x)} \frac{d^2 \psi(x)}{dx^2} + V(x) = \eta$$

---

- The time independent portion can now be written as

$$-\frac{\hbar^2}{2m} \frac{1}{\psi(x)} \frac{d^2 \psi(x)}{dx^2} + V(x) = E$$

where the separation constant is replaced by the total energy  $E$ .

- Therefore, the time independent part of W.E. is

$$\frac{d^2 \psi(x)}{dx^2} + \frac{2m}{\hbar^2} (E - V(x)) = 0$$

- This is called the time independent Schrödinger wave equation. The aim is to solve for the time independent wave function  $\psi(x)$  for a given potential  $V(x)$ .

# Physical meaning of wave function

## Probability density hypothesis

If the **wave function** of a particle is  $\Psi(x, t)$ , then the probability density function is given by

$$|\Psi(x, t)|^2$$



- Max Born in 1926 postulated the **interpretation** of wave function.
- Let us assume that we have solved for the time independent part of wave function. Thus,

$$\Psi(x, t) = \psi(x)e^{-j\omega t}$$

- Since  $\Psi(x, t)$  is a complex function, it cannot represent physical quantity. Its modulus squared is the **probability density** of finding particle between  $x$  and  $x + dx$  and is given by

$$|\Psi(x, t)|^2 = \Psi(x, t) \cdot \Psi^*(x, t) = \psi(x)e^{-j\omega t} \cdot \psi^*(x)e^{+j\omega t} = |\psi(x)|^2$$

- The probability of finding particle between  $x$  and  $x + dx$  is then

$$|\psi(x)|^2 \cdot dx$$



## Boundary conditions

- Total probability is unity.

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$$

This condition is called **normality condition**.

- Continuity condition
  1.  $\psi(x)$  must be finite, single-valued and continuous.
  2.  $\frac{\partial \psi(x)}{\partial x}$  must be finite, single-valued and continuous.
- $\psi(x)$  must vanish at infinity.

$$\lim_{x \rightarrow \infty} \psi(x) = 0, \quad \lim_{x \rightarrow -\infty} \psi(x) = 0$$

This condition is due to the **physicality condition**.

### Problem

*Normalization of wave function*

1. Consider the wave function  $\Psi(x, t) = A \cos\left(\frac{\pi x}{2}\right) e^{-j\omega t}$  for  $-1 \leq x \leq +3$ . Determine  $A$  so that  $\int_{-1}^{+3} |\Psi(x, t)|^2 dx = 1$ .

# Summary: Postulates of quantum mechanics



The work of Planck, de Broglie, Heisenberg, Born, and Schrödinger can be summarized as

1. The state of a particle is given by the wave function  $\Psi(x, t)$ .
2. The probability density of the particle is  $|\Psi(x, t)|^2$ .  
The probability of finding the particle in between  $x$  and  $x + dx$  is  $|\Psi(x, t)|^2 dx$ .
3. If the uncertainty in position is  $\Delta x$  and uncertainty in position is  $\Delta p$ , then  $\Delta x \Delta p \geq \frac{\hbar}{2}$ .
4. The de Broglie wavelength of particle with momentum  $p$  is  $\lambda = \frac{h}{p}$ .
5. The energy  $E$  of the particle with frequency  $\nu$  is  $E = h\nu$ .



## Particle in a 1D box

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# Applications of Schrödinger's W.E.

$$\frac{d^2\psi(x)}{dx^2} + \frac{2m}{\hbar^2}(E - V(x))\psi(x) = 0 \quad \Psi(x, t) = \psi(x)e^{-j\omega t}$$

- The aim is to solve for the time independent wave function  $\psi(x)$  for a given potential  $V(x)$ .
- The potential functions that are useful to describe commonly occurring physical systems are

| $V(x)$                                                                                          | Name                    | Physical systems       |
|-------------------------------------------------------------------------------------------------|-------------------------|------------------------|
| 0                                                                                               | Constant potential      | Particle in free space |
| $V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & \text{otherwise} \end{cases}$                   | Infinite potential well | Ideal atom             |
| $V(x) = \begin{cases} 0 & -\frac{a}{2} < x < \frac{a}{2} \\ V_0 & \text{otherwise} \end{cases}$ | Finite potential well   | Work function          |
| $V(x) = \begin{cases} V_0 & -\frac{a}{2} < x < \frac{a}{2} \end{cases}$                         | Finite barrier          | Nuclear fission        |

# Particle in free space

## $V(x)$ : **Constant potential**

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0 \quad \Psi(x, t) = \psi(x)e^{-j\omega t}$$

---

- Let us define the wave vector  $k$  as

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

so that the Time Independent Schrödinger Equation (TISE) is reduced to

$$\frac{d^2\psi(x)}{dx^2} + k^2\psi(x) = 0$$

so that

$$\frac{d^2\psi(x)}{dx^2} = -k^2\psi(x)$$

## $V(x)$ : Constant potential – Solution of TISE

$$\frac{d^2\psi(x)}{dx^2} = -k^2\psi(x)$$

---

- The solution of above differential equation can be written as

$$\psi(x) = Ae^{jkx} + Be^{-jkx}$$

- If we use the Euler's formula

$$e^{jx} = \cos x + j \sin x$$

then the solution can be written also as

$$\psi(x) = A' \cos kx + B' \sin kx$$



## $V(x)$ : **Constant potential – Wave vector, Momentum and Energy**

- All the wave vectors are allowed

$$k = -\infty \text{ to } +\infty$$

- The momentum is given by

$$p = \frac{h}{\lambda} \Rightarrow p = \hbar k \quad \left[ \because k = \frac{2\pi}{\lambda} \right]$$

- The corresponding energy levels are given by

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

## $V(x)$ : Constant potential – Solution of TDSE

- The solution of Time Dependent part of Schrödinger Equation (TDSE) is given by

$$\phi(t) = e^{-j\omega t}, \quad \text{where} \quad \omega = \frac{E}{\hbar} = \frac{\hbar k^2}{2m}$$

- The total wave function is given by

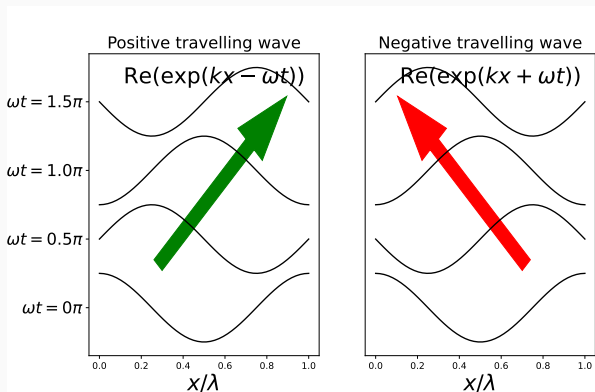
$$\Psi_k(x, t) = \psi_k(x)\phi_k(t) = e^{j\overline{kx - \omega t}}.$$

where the subscript  $k$  is the quantum number.

## $V(x)$ : Constant potential – Nature of solutions

- The solutions are **traveling** waves.
- The energy  $E$  can be any positive real number and is related to the wave vector  $k$  by

$$E = \frac{\hbar^2 k^2}{2m}$$



## Constant potential: problems

### Problem

*An electron in free space is described by a plane wave given by*

*$\Psi(x, t) = A \exp(j(kx - \omega t))$ . If  $k = 8 \times 10^8 \text{ m}^{-1}$  and*

*$\omega = 8 \times 10^{12} \text{ rad s}^{-1}$ , determine*

- 1. phase velocity and wavelength of the plane wave*
- 2. momentum and kinetic energy (in eV)*

*Repeat the steps for  $k = -1.5 \times 10^9 \text{ m}^{-1}$  and*

*$\omega = 1.5 \times 10^{13} \text{ rad s}^{-1}$ .*

### Problem

*Determine the wave number, wavelength, angular frequency, and period of wavefunction that describes an electron traveling in free space at a velocity of (a)  $5 \times 10^6 \text{ cm s}^{-1}$ .*

## Summary of particle in free space

- For a particle in free space, all wave vectors are allowed  
 $k = -\infty$  to  $+\infty$ .

- The corresponding energy levels are given by

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

- The time independent part of wave function is given by

$$\psi(x) = e^{jkx}$$

- The time dependent part of wave function is given by

$$\phi(t) = e^{-j\omega t}, \quad \text{where} \quad \omega = \frac{E}{\hbar}$$

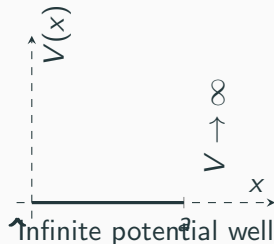
- The total wave function is given by

$$\Psi_k(x, t) = \psi_k(x)\phi_k(t) = e^{j\overline{kx - \omega t}}.$$

\*Particle in a 1D box\*

## $V(x)$ : Infinite potential

$$V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & \text{otherwise} \end{cases}$$



- Consider a particle in a 1-dimensional infinite potential well.
- The well can be considered as a box.

Outside infinite box

$$\frac{d^2\psi(x)}{dx^2} + \frac{2m}{\hbar^2}(E - \infty)\psi(x) = 0$$

Inside infinite box

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0$$

## $V(x)$ : Infinite potential well – Solution of TISE

Outside infinite box

- There is no wave function!

$$\psi(x) = 0$$

- Upon applying boundary conditions

Inside infinite box

- The wavefunction is

$$\psi(x) = A \cos kx + B \sin kx$$

Left B.C.

$$\psi(x=0) = 0$$

$$A \cos kx = 0$$

$$\Rightarrow A = 0$$

Right B.C.

$$\psi(x=a) = 0$$

$$B \sin ka = 0$$

$$\Rightarrow \sin ka = 0$$



## $V(x)$ : Infinite potential well – Solution of TISE

- The equation  $\sin ka = 0$  is valid if

$$ka = n\pi$$

where  $n$  is a positive integer, or  $n = 1, 2, 3, \dots$ . The parameter is referred to as quantum number. In terms of the quantum number, the wave vector is given by

$$k = \frac{n\pi}{a}$$

- The prefactor  $B$  can be found by applying the **normality condition** of wave function.

$$\int_0^a |B \sin kx|^2 dx = 1, \quad \Rightarrow \quad \int_0^a B^2 \sin^2 kx dx = 1$$

$$\therefore B = \sqrt{\frac{2}{a}}$$

- Therefore, the time independent wave function is given by

$$\psi(x) = \sqrt{\frac{2}{a}} \sin(n\pi x/a)$$

## $V(x)$ : Infinite potential well – Energy levels

- Recall that the wave vector  $k$  is related to the energy  $E$  by

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \quad \Rightarrow \quad E = \frac{\hbar^2 k^2}{2m}$$

- Therefore, the energy is given by

$$E = E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2}$$

### Problem

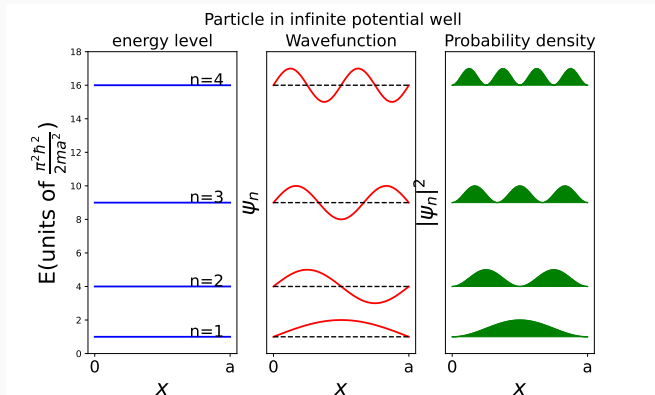
*Electron in an infinite potential well of width  $5 \text{ \AA}$ . Calculate the first three energy levels.*

### Key Insight

The energy of particle is **quantized** i.e. it can only have particular **discrete** values. 

## $V(x)$ : Infinite potential well – Nature of solutions

- The solutions are **standing** waves.



# Orthogonality condition

- Wave function associated with an energy level is normalized.
- In addition to normality, wave functions  $\psi_m(x), \psi_n(x)$  corresponding to any two levels  $m, n$  of infinite potential well are orthogonal to each other i.e.

$$\int_0^a \psi_m^*(x) \psi_n(x) dx = 0.$$

This is called the **orthogonality condition**.

## Theorem

*Prove the orthogonality condition for solutions of infinite potential well.*

# 1D infinite well: problems

## Problem

1. *A particle with mass of 15 mg is bound in a 1D infinite potential well that is 1.2 cm wide.*
2. *If the energy of the particle is 15 mJ, determine the value of  $n$  for that state.*
3. *What is the energy of the  $(n + 1)$  state?*
4. *Would quantum effects be observable for this particle?*

## Summary of particle in infinite potential well

- For a particle in an infinite potential well, discrete wave vectors are allowed and are given by

$$k_n = \frac{n\pi}{a}.$$

- The corresponding energy levels are given by

$$E_n = \frac{\hbar^2 k^2}{2m}, \quad \text{or} \quad E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2}$$

- The time independent part of wave function is given by

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

- The time dependent part of wave function is given by

$$\phi_n(t) = e^{-j\omega_n t}, \quad \text{where} \quad \omega_n = \frac{E_n}{\hbar}$$

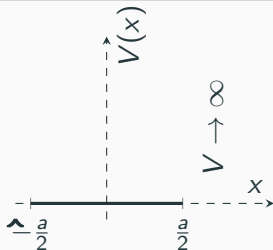
- The total wave function is given by

$$\Psi_n(x, t) = \psi_n(x)\phi_n(t),$$

\*Particle in a symmetric form infinite  
potential well\*

## $V(x)$ : Symmetric Infinite potential

$$V(x) = \begin{cases} 0 & -\frac{a}{2} \leq x \leq +\frac{a}{2} \\ \infty & \text{otherwise} \end{cases}$$



- Consider a particle in a **symmetric form** of 1-dimensional infinite potential well.
- The analysis for this potential is similar to the analysis for the asymmetric form discussed earlier.
- However, the general form of wave function can be considered as linear combination of exponential functions instead of trigonometric functions due to the symmetry of the problem.
- Thus, let the wave function inside the well be

$$\psi(x) = Ae^{jkx} + Be^{-jkx}$$



## $V(x)$ : Symmetric Infinite potential – Solution of TISE

$$\psi(x) = Ae^{jkx} + Be^{-jkx}$$

---

- Upon applying left boundary condition,

$$\psi(x = -\frac{a}{2}) = 0$$

$$\Leftrightarrow A \exp\left(-j\frac{ka}{2}\right) + B \exp\left(+j\frac{ka}{2}\right) = 0$$

$$\Leftrightarrow (A + B) \cos\left(\frac{ka}{2}\right) + j(-A + B) \sin\left(\frac{ka}{2}\right) = 0$$

$$\Leftrightarrow (A + B) \cos\left(\frac{ka}{2}\right) = 0 \text{ and } (-A + B) \sin\left(\frac{ka}{2}\right) = 0$$

$$\Leftrightarrow [A + B = 0 \text{ or } \cos\left(\frac{ka}{2}\right) = 0] \text{ and } [A - B = 0 \text{ or } \sin\left(\frac{ka}{2}\right) = 0]$$

$$\Leftrightarrow [A - B = 0 \text{ and } \cos\left(\frac{ka}{2}\right) = 0] \text{ or } [A + B = 0 \text{ and } \sin\left(\frac{ka}{2}\right) = 0]$$

- Same cases arise for the right boundary condition due to **symmetry** of the problem!

## $V(x)$ : Symmetric Infinite potential – Solution of TISE

$$\psi(x) = Ae^{jkx} + Be^{-jkx}$$

---

Case (i) Even wave functions

$$A - B = 0$$

and

$$\cos\left(\frac{ka}{2}\right) = 0$$

$$\Leftrightarrow \psi(x) = A \cos kx$$

$$\Leftrightarrow \frac{ka}{2} = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$$\Leftrightarrow k = \frac{\pi}{a}, \frac{3\pi}{a}, \dots$$

Case (ii) Odd wave functions

$$A + B = 0$$

and

$$\sin\left(\frac{ka}{2}\right) = 0$$

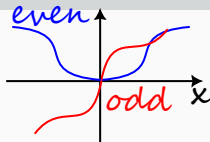
$$\Leftrightarrow \psi(x) = A \sin kx$$

$$\Leftrightarrow \frac{ka}{2} = \pi, 2\pi, \dots$$

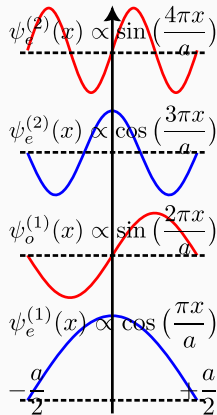
$$\Leftrightarrow k = \frac{2\pi}{a}, \frac{4\pi}{a}, \dots$$

## $V(x)$ : Symmetric Infinite potential – Nature of solutions

$$f(x) := \begin{cases} \text{even} & \text{if } f(-x) = f(x) \\ \text{odd} & \text{if } f(-x) = -f(x) \end{cases}$$



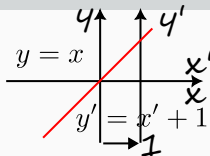
- The potential function is an even function as  $V(-x) = V(x)$ .
- The wave functions belonging to first case are even functions as  $\cos x$  is an even function whereas the wave functions belonging to second case are odd functions as  $\sin x$  is an odd function.
- There are two subfamilies of even and odd wave functions
- If we order the wave functions with increasing wave vector, then wave



## $V(x)$ : Symmetric Infinite potential – Alternative solution

- Translation of origin by  $\alpha$  transforms the function in new coordinate frame

$$f^{\text{new}}(x) = f^{\text{old}}(x + \alpha)$$



- The solutions to the symmetric form of infinite potential well can be obtained from the solutions to the asymmetric form by translating the coordinate frame by  $\frac{a}{2}$ .

$$x \rightarrow x + \frac{a}{2}$$

- The wave functions of the symmetric potential well are given by

$$\psi^{\text{sym}}(x) = \psi^{\text{asym}}(x + \frac{a}{2})$$

$$\psi^{\text{sym}}(x) = \psi^{\text{asym}}(x + \frac{a}{2})$$

$$= \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x + \frac{n\pi}{2}\right) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a} + \frac{n\pi}{2}\right)$$

## $V(x)$ : Symmetric Infinite potential – Alternative solution

- Upon substituting for  $n = 1, 2, \dots$ , we have

$$\psi^{\text{sym}}(x) \propto \begin{cases} \sin\left(\frac{\pi x}{a} + \frac{\pi}{2}\right) & n = 1 \\ \sin\left(\frac{2\pi x}{a} + \frac{2\pi}{2}\right) & n = 2 \\ \sin\left(\frac{3\pi x}{a} + \frac{3\pi}{2}\right) & n = 3 \\ \sin\left(\frac{4\pi x}{a} + \frac{4\pi}{2}\right) & n = 4 \\ \vdots & \vdots \end{cases}$$

- Using the trigonometric relations, we have

$$\psi^{\text{sym}}(x) \propto \begin{cases} \cos\left(\frac{\pi x}{a}\right) & n = 1 \\ \sin\left(\frac{2\pi x}{a}\right) & n = 2 \\ \cos\left(\frac{3\pi x}{a}\right) & n = 3 \\ \sin\left(\frac{4\pi x}{a}\right) & n = 4 \\ \vdots & \vdots \end{cases}$$

In some cases, prefactor of  $-1$  can be absorbed into normalizing factor.

## Summary of particle in symmetric infinite potential well

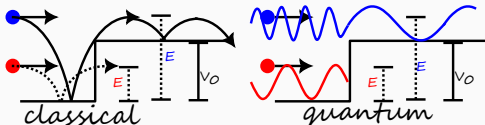
- For a particle in a symmetric infinite potential well, the allowed wave vectors are same as that of asymmetric infinite potential well.
- The corresponding energy levels are also same
- The potential well function is an even function i.e.

$$V(-x) = V(x).$$

Due to the symmetry of the potential well function, the wave functions are either **odd** or **even**.

\*Particle in a finite potential well\*

# Potential energy barrier – Classical vs Quantum



- There are two regions – (I) without barrier and (II) with barrier

## Region I

- Total energy  $E$  is kinetic energy  $T_1$
- Wave vector is real
- Wave function has sinusoidal form

$$\psi_1(x) = A \exp jk_1x + B \exp -jk_1x$$

$$T_1 = E$$

$$k_1 = \sqrt{\frac{2mT_1}{\hbar^2}}$$

## Region II

- There are two cases – (i) total energy is greater than potential



## Case (i) – $E > V_0$

- Classically, the particle overcomes the potential barrier with decreased kinetic energy.
- Quantum mechanically, the particle overcomes the potential barrier with decreased wave vector.

### Region II

- Kinetic energy is reduced by  $V_0$
- The wave function is given by

$$T_{II} = E - V_0$$

$$\frac{d^2\psi(x)}{dx^2} + \frac{2m(E - V_0)}{\hbar^2}\psi(x) = 0$$

- Wave vector is real

$$k_{II} = \sqrt{\frac{2m(E - V_0)}{\hbar^2}}$$

- Wave vector is reduced compared to region I

## Curious case – $E < V_0$

- 

Classically, the particle **cannot** overcome the potential barrier.

- However, quantum mechanically, it is possible!

- Negative kinetic energy is allowed in quantum mechanics.

### Region II

- Total energy  $E$  is less than potential barrier

- Kinetic energy is negative!

$$T_{II} = E - V_0 < 0$$

- Wavevector is imaginary

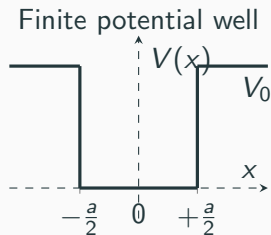
$$k_{II} = \sqrt{\frac{2m(E - V_0)}{\hbar^2}} = j\sqrt{\frac{2m(V_0 - E)}{\hbar^2}} = j\kappa_{II}$$

- Wave function has exponential form

$$\psi_{II}(x) = A \exp jk_{II}x + B \exp -jk_{II}x$$

## $V(x)$ : Finite potential well

$$V(x) = \begin{cases} 0 & -\frac{a}{2} < x < \frac{a}{2} \\ V_0 & \text{otherwise} \end{cases}$$



- There are three regions – (I) well, (II) right of well, and (III) left of well.
- There are two cases – (i) total energy is less than potential barrier and (ii) total energy is greater than potential barrier
- Let us solve for case (i).

## $V(x)$ : Finite potential well – Case (i): $E < V_0$

### Region I

- Kinetic energy is positive so

$$\psi_I(x) = A \exp(jk_I x) + B \exp(-jk_I x)$$

- Apply the **symmetry condition** of even or odd wave function

$$\text{Case (1): } A = B \quad \text{or} \quad \text{Case (2): } A = -B$$

### Region II

- Kinetic energy is negative so

$$\psi_{II}(x) = C \exp(-\kappa_{II} x) + D \exp(+\kappa_{II} x)$$

- Apply the **finiteness condition**

$$\lim_{x \rightarrow \infty} \psi_{II}(x) = \text{finite}$$

$$\Leftrightarrow D = 0.$$

### Region III

- Kinetic energy is negative so

$$\psi_{III}(x) = E \exp(-\kappa_{III} x) + F \exp(\kappa_{III} x)$$

- Apply the **finiteness condition**

$$\lim_{x \rightarrow -\infty} \psi_{III}(x) = \text{finite}$$

## $V(x)$ : Finite potential well – case (1): $A = B$

- Interface of Region I and Region II i.e.  $x = \pm \frac{a}{2}$
- 

Continuity condition of  $\psi$

$$\begin{aligned}\psi_I\left(+\frac{a}{2}\right) &= \psi_{II}\left(+\frac{a}{2}\right) \\ \Leftrightarrow A \left[ \exp\left(j\frac{k_I a}{2}\right) + \exp\left(-j\frac{k_I a}{2}\right) \right] &= C \exp\left(-\frac{\kappa_{II} a}{2}\right) \\ \Leftrightarrow 2A \cos \frac{k_I a}{2} &= C \exp -\frac{\kappa_{II} a}{2} \quad (1)\end{aligned}$$

Continuity condition of  $\frac{d\psi}{dx}$

$$\begin{aligned}\frac{d\psi_I}{dx}\left(+\frac{a}{2}\right) &= \frac{d\psi_{II}}{dx}\left(+\frac{a}{2}\right) \\ \Leftrightarrow jAk_I \left[ \exp\left(j\frac{k_I a}{2}\right) - \exp\left(-j\frac{k_I a}{2}\right) \right] &= -C\kappa_{II} \exp -\frac{\kappa_{II} a}{2} \\ \Leftrightarrow 2Ak_I \sin \frac{k_I a}{2} &= C\kappa_{II} \exp -\frac{\kappa_{II} a}{2} \quad (2)\end{aligned}$$

- Dividing (1) by (2) gives the **quantization condition** for wave vector.

$$\tan \frac{k_I a}{2} = \frac{\kappa_{II}}{k_I}$$

## $V(x)$ : Finite potential well – case (1): $A = B$

$$\tan \frac{k_I a}{2} = \frac{\kappa_{II}}{k_I} \quad (3)$$

- This is a transcendental type of equation <sup>3</sup> and **cannot** be solved analytically!
- We have to employ numerical techniques.
- First let us introduce dimensionless variable  $z$  and constant  $z_0$

$$z = \frac{k_I a}{2} = \frac{a}{2} \sqrt{\frac{2mE}{\hbar^2}}, \quad z_0 = \frac{a}{2} \sqrt{\frac{2mV_0}{\hbar^2}}$$

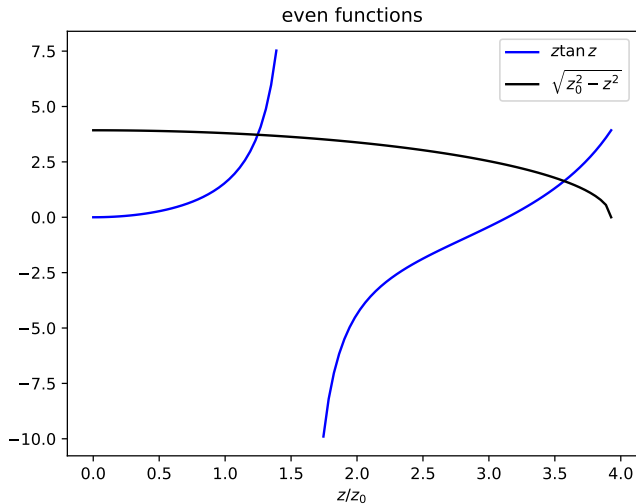
so that (3) is reduced to

$$z \tan z = \sqrt{z_0^2 - z^2}.$$

- Now let us plot the graph of LHS and RHS of above equation. The points of intersection given you quantized values of  $z$ .

From this, we can get quantized wave vectors and quantized <sup>72 / 76</sup>

## $V(x)$ : Finite potential well – case (1): $A = B$



## $V(x)$ : **Finite potential well – case (1):** $A = B$

- By symmetry the other interface between Region I and Region III gives the same quantization condition.
- Let us denote the quantized wave vectors as

$$k_I^{(1)}, k_I^{(2)}, k_I^{(3)}, \dots, \quad \text{and} \quad \kappa_{II}^{(1)}, \kappa_{II}^{(2)}, \kappa_{II}^{(3)}, \dots$$

- By symmetry  $F = C$ .
- $C$  can be expressed in terms of  $A$  from (1).
- Therefore, the wave functions are given by

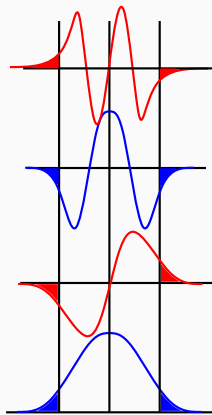
$$\psi^{(n)}(x) = \begin{cases} C \exp\left(+\kappa_{II}^{(n)} x\right) & x < -\frac{a}{2} \\ 2A \cos \frac{k_I^{(n)} x}{2} & -\frac{a}{2} < x < +\frac{a}{2} \\ C \exp\left(-\kappa_{II}^{(n)} x\right) & x > +\frac{a}{2} \end{cases}$$

- The solutions corresponding to  $E < V_0$  are called **bound states**.



## $V(x)$ : Finite potential well – Nature of solutions

- Wave function penetrates into the finite potential barrier.
- The wave function is either odd or even function.
- The depth of the well determines the number of bound states.
- In the limiting case of very large depth, the penetration tends to zero and the number of bound states tends to infinity leading to particle in infinite potential well.



### Key Insight

The wave function penetrates into the finite potential barrier. This is called **Quantum Penetration**. 

## $V(x)$ : Finite potential well – case (2): $A = -B$

- Similar analysis for the case(2):  $A = -B$  gives the quantization condition as

$$\cot \frac{k_1 a}{2} = -\frac{\kappa_{II}}{k_1}$$

### **Problem**

*Prove the above quantization condition for the odd wave functions.*

# Quantum dots

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# Finite Potential well – Quantum tunneling

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